

Hedge Funds

Chapter Summary

Hedge funds (HFs) are not an asset class. HF returns have large exposure to dynamic factors—especially volatility risk. After taking these nonlinear risks into account, the average HF is unlikely to add value. HF fees are high, but, contrary to popular perception, only a minority of HF manager compensation comes from incentive fees.

1. The Quant Meltdown

In the first week of August 2007, returns of quantitative hedge funds (quant HFs) went into a tailspin.¹ From August 7 to 9, quant funds experienced huge and unprecedented losses—for no obvious reason. There was no overall market decline, and the large losses occurred only at quant funds.

Quant investing picks securities by mining data and creating quantitative signals. Its development dates to the 1960s, when the new Capital Asset Pricing Model (see chapter 6) gave fund managers a way to estimate expected returns of stocks from beta—a statistic measuring how a stock moves together with the market. High-frequency statistical arbitrage funds use highly technical strategies with very short horizons (ranging from seconds to one or two days). Longer-term *market neutral* funds have investment horizons from weeks to months and rely on economic models and lower-frequency statistical forecasting methods. Some market-neutral funds tend to use factor-based investment strategies like value-growth, size, momentum, volatility, and credit (see chapter 7). Many HFs are leveraged and take short positions.

The complex models used by quant funds amplified the funds' decline during those three grim August days. Amir Khandani, a graduate student at MIT, and his

¹ This is based on "The Quant Meltdown: August 2007," CaseWorks ID #080317. See also Khandali and Lo (2007) and Daniel (2009).